KARTHIK KALIDAS

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PROFESSIONAL SUMMARY

Technical leader with 7 years of experience delivering autonomous vehicle platforms from concept to deployment. Expert in HIL infrastructure, sensor integration, and large-scale test and metrics automation. Proven ability to execute and scale solutions, solve complex problems, and drive cross-functional team success.

WORK EXPERIENCE

Staff Systems Engineer, HIL Validation & Release Metrics — Apr 2026 – Present
Torc Robotics, Fort Worth, TX

- Accountable for Torc's Metrics computation pipeline with 20+ performance and safety metrics across HIL, simulation, and on-road data to govern executive-level decisions for L4 software
- Directing HIL roadmap to institutionalize shift-left validation across component, subsystem, and full-system tiers

Senior Systems Engineer, HIL Validation — Jan 2025 – Mar 2026
Torc Robotics, Fort Worth, TX

- Spearheading HIL validation roadmap across 5 autonomy teams, boosting system test coverage by 70%
- Deployed fully automated robotic HIL test via CI/CD using temporal workflows, achieving zero-touch software flashing, mission validation, and mode engagement in under 20 mins
- Boosted HIL stability and uptime from 30% to 99% through proactive monitoring and redundancy solutions

System Integration Engineer, Autonomous Driving Kit — Nov 2021 – Jan 2025
Torc Robotics, Albuquerque, NM

- Led rapid integration and deployment of 4 vehicle generations in collaboration with Daimler Truck for mass production
- Developed patented sensor setup station for full sensor set configuration in <12 mins, utilized on over 25 trucks

Test Engineer, On-Road Testing — Jan 2021 – Nov 2021
Torc Robotics, Albuquerque, NM

- Designed extrinsic calibration procedure for lidars, radars, and cameras using internal tooling
- Led establishment of Torc's first semi-truck test operations facility, building and training the test engineering team

PUBLICATIONS

2025 - Systems And Methods For Automatic Sensor Configuration, 139906-05801
2025 - A Cloud-Based Platform For Automatic Sensor Configuration, 139906-08501
2021 - Simulation Framework for Testing Autonomous Vehicles, SAE Technical Paper 2021-01-0118

SKILLS

Languages	Python, C/C++17, Go, Bash, Vue.js
Tools/Framework	ROS1/2, Bazel, MATLAB, CANape, SolidWorks, Jama, Cameo
DevOps	Linux, GitHub Actions, Jenkins, Terraform, Docker, AWS, Datadog

AWARDS

2024 – Employee of the Quarter Q2, Torc Robotics
2021 – Employee of the Quarter Q1, Torc Robotics
2019 – Formula Student Award, IIT Bombay Racing
2018 – Institute Technical Citation, IIT Bombay
2015 - All India Rank 606, JEE Advanced

EDUCATION

The Ohio State University, Columbus, OH — Dec 2020
Master of Science, Mechanical Engineering — GPA: 4.0/4.0

Indian Institute of Technology (IIT) Bombay, India — Aug 2019
Bachelor of Technology, Mechanical Engineering — CPI: 8.4/10.0